

StatsBomb Open Data

3,961 matches · 24 competitions · men's & women's football
event streams + 360 freeze-frames

E1 · Passing-style & event-frequency features

15 informative (3 pass-space + 12 event)

PCA → k-means

11 comps · 95.9% var · k=3 · sil 0.134

Tactical archetypes

C0 transition · C1 turnover · C2 possession

E2 · Freeze-frame spatial features

defensive structure · pressing · lanes

Complementarity vs event streams

mean $|r| \leq 0.20$ · clustering ARI ≈ 0

Independent spatial axis

orthogonal to event-stream tactics

E3 · Passing-network graph

20 players · 16-d node features

→ GCN outcome model (Fig. 2)

graph representation of tactics

Outcome prediction (W / D / L)

see Results §4.3

Cross-domain validation

gender transfer: male → female 0.544

league transfer: La Liga 0.528 · Ligue 1 0.509 · Serie A 0.495 · PL 0.491

→ **transferable tactical structure: no consistent degradation**